

Problem Statement: 2D Geometry Calculator Using Python Modules

Objective

Develop a menu-driven Python application that demonstrates the use of **modules, functions, loops, and conditional statements** by calculating the **Area** and **Perimeter** of different two-dimensional figures.

Problem Description

A geometry learning application requires a utility to perform calculations related to various 2D figures. Your task is to design this application in Python using a **modular programming approach**.

Part 1: Create a Python Module

Create a Python module named **geometry.py** that contains separate functions to perform the following operations:

Circle

- Calculate the area of a circle.
- Calculate the perimeter (circumference) of a circle.

Square

- Calculate the area of a square.
- Calculate the perimeter of a square.

Rectangle

- Calculate the area of a rectangle.
- Calculate the perimeter of a rectangle.

Note: Use appropriate parameters for each function. For example, a circle requires a radius, a square requires a side length, and a rectangle requires length and breadth.

Part 2: Develop the Main Application

Create another Python program named **main.py** that imports the functions from the **geometry.py** module and provides the following functionality:

Step 1: Display Figure Menu

Prompt the user to select one of the following 2D figures:

1. Circle
 2. Square
 3. Rectangle
 4. Exit
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Step 2: Accept Required Inputs

Based on the selected figure, ask the user to enter the required dimensions.

Examples:

- Circle → Radius
 - Square → Side
 - Rectangle → Length and Breadth
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Step 3: Display Operation Menu

After collecting the input values, display another menu to allow the user to select the desired operation:

1. Area Calculation
2. Perimeter Calculation
3. Change Figure

Perform the selected operation by invoking the appropriate function from the module and display the result.

Step 4: Repeat Operations on the Same Figure

After displaying the result, ask the user whether they want to perform another operation on the **same figure**.

Example:

Do you want to perform another operation on the same figure? (Y/N):

- If the user enters **Y**, display the operation menu again.
 - If the user enters **N**, proceed to the next step.
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Step 5: Continue or Exit the Application

After exiting from the current figure, ask the user:

Do you want to continue using the application? (Y/N):

- If the user enters **Y**, display the figure selection menu again.
 - If the user enters **N**, terminate the program with an appropriate message.
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Additional Requirements

1. Use **functions** to improve code readability and maintainability.
 2. Use a **separate Python module** for all geometry-related calculations.
 3. Implement **input validation** to ensure that dimensions entered by the user are positive numbers.
 4. Use **nested loops** appropriately to support repeated operations and program continuation.
 5. Display meaningful messages for invalid menu choices.
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Sample Execution Flow

Select the Figure:

1. Circle
2. Square
3. Rectangle
4. Exit

Enter your choice: 1

Enter radius: 5

Select Operation:

1. Area
2. Perimeter
3. Change Figure

Enter your choice: 1

Area of Circle = 78.54

Do you want to perform another operation on the same figure? (Y/N): Y

Select Operation:

1. Area
2. Perimeter
3. Change Figure

Enter your choice: 2

Perimeter of Circle = 31.42

Do you want to perform another operation on the same figure? (Y/N): N

Do you want to continue using the application? (Y/N): Y
